The University of Jordan

School of Engineering

Department of Computer Engineering

Practical Numerical Analysis (CPE313)

Labsheet 09 – Differentiation and Integration

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Exercise 1 – Centered Difference Approximation of the First Derivative (8 Marks)

You are required to write a script file that finds the derivative of x^2e^{-x} at $x = 1.25$ within an $\varepsilon_s = 0.1\%$ centered difference approximation.

You need to find Δx that satisfies the required error difference (start with $\Delta x = 2$, compute the centered difference approximation, calculate ε_a and in each iteration the Δx is half the previous one).

```
y = @(x) x.^2 .* exp(-x);
y_prime = @(x) (-x.^2 .* exp(-x) + 2 .* x .* exp(-x));
true_value = y_prime(1.25);
delta_x = 2;
eps = Inf;
```

```
while abs(eps) > 0.1

    derivative = (y(1.25 + delta_x) - y(1.25 - delta_x)) / (2 * delta_x);

    eps = ((true_value - derivative) / true_value) * 100;

    delta_x = delta_x / 2;
end

disp([' delta_x: ', num2str(delta_x)]);
```

```
delta_x: 0.0625
```

```
disp([' derivative: ', num2str(derivative)]);
```

```
derivative: 0.26855
```

```
disp([' error: ', num2str(abs(eps))]);
```

```
error: 0.01933
```

Exercise 2 – Simspons 1/3 Rule (7 Marks)

You are required to write a MATLAB code that calculates the area under the curve for any given function f(x) using Simpson's 1/3 rule.

Your script must be initialized with:

- The function to be integrated: $f(x) = 4 - 2x + 0.375x^3$
- An integration interval between -2 and 7
- The number of segments **n** that you want to use for your integral approximation. Try your code with n = 10, and n = 20.

```
f = @(x) 4 - 2.*x + 0.375.*x.^3;
a = -2; b = 7;

for n = [10, 20]
    h = (b - a) / n;
    x = linspace(a, b, n+1);
    y = f(x);

    integral_area = y(1) + y(end);
    integral_area = integral_area + 4 * sum(y(2:2:end-1));
    integral_area = integral_area + 2 * sum(y(3:2:end-2));
    integral_area = integral_area * h / 3;
```

```
    fprintf('For n = %d, Area = %.6f\n', n, integral_area);
end
```

```
For n = 10, Area = 214.593750
For n = 20, Area = 214.593750
```

Exercise 3 – MATLAB Integration (3 Marks)

Use the two MATLAB's built-in functions to integrate the same function above and compare your results.

```
x = linspace(-2, 7, 10000)
```

```
x = 1×10000
   -2.0000   -1.9991   -1.9982   -1.9973   -1.9964   -1.9955   -1.9946   -1.9937 ···
```

```
y = 4 - 2 .* x + 0.375 .* x .^ 3
```

```
y = 1×10000
    5.0000    5.0022    5.0045    5.0067    5.0090    5.0112    5.0134    5.0157 ···
```

```
t = trapz(x,y)
```

```
t =
214.5938
```

```
y = @(x) 4 - 2 .* x + 0.375 .* x .^ 3
```

```
y = function_handle with value:
    @(x)4-2.*x+0.375.*x.^3
```

```
q = integral(y,-2,7)
```

```
q =
214.5937
```


Labsheet version 2.0

Original Labsheet October 5th, 2020

Last Updated Feb 21st, 2024

Revision History

Ver 1.2 (Feb 21st, 2024)

- Changed the format from Word File to MATLAB Live Script to make it easier for students and grader.
- Changed the functions and numbers in the experiment.
- Switched to centered difference approximation instead of backward difference approximation.
- Changed the integration part to a script rather than a function

Ver 1.1

- Moved sum of the exercises from this experiment to previous ones after expanding the numerical material from 3 to 4 experiments

Ver 1.0

- Original version